

According to Darcy's law,

$$Q = K i A$$

For radial flow towards a well, the area through which water flows at radius r is: $A = 2\pi r b$

The hydraulic gradient is: $i = -dh/dr$

Therefore,

$$Q = K \times (-dh/dr) \times 2\pi r b$$

$$Q = -2\pi K b r (dh/dr)$$

Rearranging,

$$Q/(2\pi K b) = -r(dh/dr)$$

Therefore,

$$(Q/2\pi K b) dr/r = -dh$$

Integrating between r_1 and r_2 :

$$\int (Q/2\pi K b) (dr/r) = -\int dh$$

$$(Q/2\pi K b) \ln(r_2/r_1) = h_1 - h_2$$

Therefore,

$$Q = \frac{2\pi K b (h_1 - h_2)}{\ln(r_2/r_1)}$$

Hence, the discharge from a confined aquifer is:

$$Q = \frac{2\pi K b (h_1 - h_2)}{\ln(r_2/r_1)}$$

Or, using common logarithm:

$$Q = \frac{2.303 \times 2\pi K b (h_1 - h_2)}{\log(r_2/r_1)}$$

This equation is known as the Thiem equation for steady radial flow in a confined aquifer.

2. **What are the principal sources of water pollution? Explain their effects on the quality of water.**

Answer:

Water pollution also known as aquatic pollution is the contamination of water bodies which alters the physical, chemical and biological composition of water. Bagmati, Bishnumati, Dhobi Khola, Hanumante and Manohara rivers in Kathmandu Valley are among the most polluted water bodies because of sewage, solid waste and industrial discharge.

The principal sources of water pollution are as follows:

- i. Sewage and wastewater
- ii. Household detergents and chemicals

- iii. Industrial effluents, Toxic chemicals and dyes
- iv. Heavy metals (lead, mercury, cadmium)
- v. Fertilizers (nitrates and phosphates)
- vi. Pesticides and herbicides
- vii. Animal waste
- viii. Oil spills and landfill leachate
- ix. Mining activities
- x. Hospital and pharmaceutical waste
- xi. Urban stormwater runoff

The effects in the quality of water are listed below:

- i. **Depletion of Dissolved Oxygen (DO):**
Domestic sewage is rich in organic matter. Microorganisms consume the water's dissolved oxygen to decompose this matter, which increases the Biochemical Oxygen Demand (BOD) and rapidly decreases the DO content.
- ii. **Destruction of Aquatic Life:**
Aquatic ecosystems rely on sufficient DO for survival. Severe oxygen depletion caused by high BOD can lead to mass "fish kills" and the total destruction of aquatic habitats.
- iii. **Spread of Diseases:**
Polluted water becomes a breeding ground for pathogens. When used for drinking, bathing, or food preparation, it can spread waterborne epidemics such as typhoid, cholera, and dysentery among both humans and cattle.
- iv. **Septic Conditions and Foul Odors:**
Excessive organic matter creates anaerobic conditions, leading to the production of gases like hydrogen sulphide, methane, and ammonia. This makes the water unpleasant, foul-smelling, and unsuitable for any use.
- v. **Physical Deterioration:**
Pollution increases turbidity (cloudiness) and alters the colour and taste of the water. High sediment loads from landslides or erosion further increase turbidity, which can damage aquatic ecosystems and clog intake structures.
- vi. **Chemical Contamination:**
The presence of heavy metals and toxic industrial chemicals makes water poisonous and long-lasting in the environment, posing chronic health risks.
- vii. **Ecological Imbalance:**
High nutrient levels from fertilizers lead to excessive microbial and bacterial growth, which disturbs the natural ecological balance.

3. **A village of Nepal with design year pollution of 500 has water demand of 65 liters/capita/day. The demand is to be met by a continuous system of supply from a spring source with a safe yield of 0.3 lps. The consumption pattern of village is as follows:**
Is a balancing reservoir necessary? Calculate the capacity if necessary.

Time (hr)	Consumption pattern (%)
5:00-7:00	25
7:00-12:00	35
12:00-17:00	15
17:00-19:00	20
19:00-5:00	5

Answer:

Given:

Population, $P = 500$

Per capita water demand = 65 litres/capita/day

Safe yield of spring source = 0.3 L/s

The water consumption pattern is:

Time	Duration (hr)	Consumption (%)
5:00–7:00	2	25
7:00–12:00	5	35
12:00–17:00	5	15
17:00–19:00	2	20
19:00–5:00	10	5
Total	24	100

$$Q_d = P \times \text{Per capita demand}$$

$$= 500 \times 65$$

$$= 32,500 \text{ L/day}$$

Average discharge required is:

$$Q_{\text{avg}} = \frac{32500}{24 \times 60 \times 60}$$

$$= 0.376 \text{ L/s}$$

Now, the safe yield of spring:

$$Q_s = 0.3 \times 24 \times 60 \times 60$$

$$= 25920 \text{ L/d}$$

But the daily demand is

$$32,500 \text{ L/day} > 25,920 \text{ L/day}$$

Hence,

$$\text{Daily demand} > \text{Daily supply}$$

$$\text{The source has a deficit of: } 32,500 - 25,920 = 6,580 \text{ L/day}$$

Hence, balancing reservoir is required. However, providing balancing reservoir alone cannot solve the problem.

4. Sketch typical layout of water supply schemes in

i) Hilly zone of Nepal

ii) Terai zone of Nepal

Discuss in what cases you design following structures in water supply projects.

i) Sedimentary tank

ii) Filtration unit

iii) Aeration chamber

Answer:

The typical layout of water supply schemes in hilly zone of Nepal:

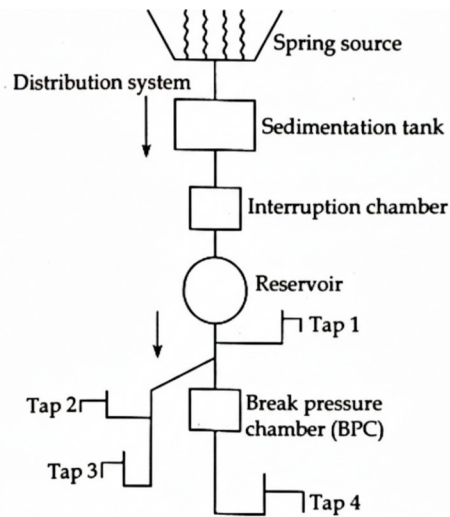


Figure: Typical rural water supply system

The typical layout of water supply schemes in terai zone of Nepal:

i. Sedimentation tank:

It is provided when the raw water contains a considerable amount of suspended and settleable impurities that can be removed by gravity. It is designed in the following cases:

- When water contains a high concentration of suspended solids, silt, sand and clay.
- When the turbidity of raw water is high, particularly during the rainy season.
- When the water contains particles that are heavier than water and capable of settling.
- To reduce the load on subsequent filtration units.
- When coagulation is used, a sedimentation tank is provided to remove the flocculated particles.

Example: Surface water from rivers and streams generally requires sedimentation before filtration.

ii. Filtration unit:

It is provided when water requires further removal of fine suspended particles and microorganisms after sedimentation. It is designed in the following cases:

- When fine suspended particles cannot be removed effectively by sedimentation.
- To remove remaining turbidity and colloidal particles.
- To remove a significant portion of bacteria and other microorganisms.
- When the treated water must satisfy the required drinking-water quality standards.
- As a polishing treatment after sedimentation or coagulation-flocculation.

Thus, filtration is generally provided after sedimentation in a conventional surface-water treatment plant.

Example: River water containing fine suspended matter and microorganisms normally requires filtration after sedimentation.

iii. Aeration chamber:

An aeration chamber is provided when raw water contains dissolved gases, undesirable volatile substances, or certain dissolved minerals that can be removed or oxidized by contact with air. It is designed in the following cases:

- To remove dissolved gases such as carbon dioxide (CO_2) and hydrogen sulphide (H_2S).
- To reduce bad taste and odor caused by dissolved gases.

- To increase the dissolved oxygen (DO) content of water.
- To oxidize soluble forms of iron and manganese into insoluble forms, which can subsequently be removed by sedimentation or filtration.
- To improve the overall physical and chemical quality of water.

Example: Groundwater containing excessive iron, manganese, H_2S or CO_2 may require aeration before further treatment.

5. **Calculate 5 days $20^\circ C$ BOD of a sewage of sample whose 5 days $30^\circ C$ BOD is 110 mg/liter. Assume the deoxygenation constant at $20^\circ C$, $k_{20}=0.1$.**

Answer:

Given,

5 days BOD at $30^\circ C$,

$BOD_5 (30) = 110 \text{ mg/litre}$

Deoxygenation constant at $20^\circ C$,

$k_{20} = 0.1$

Temperature coefficient,

$\theta = 1.047$

The relationship between deoxygenation constants at different temperatures is:

$$\begin{aligned} k_{30} &= k_{20} \times \theta^{(30-20)} \\ &= 0.1 \times (1.047)^{10} \\ &= 0.1583 \end{aligned}$$

The BOD equation is:

$$BOD_t = L (1 - e^{(-kt)})$$

For 5 days BOD at $30^\circ C$:

$$110 = L (1 - e^{(-0.1583 \times 5)})$$

Therefore,

$$\begin{aligned} L &= \frac{110}{1 - e^{(-0.7915)}} \\ &= \frac{110}{1 - 0.453} \\ &= 201.16 \text{ mg/litre} \end{aligned}$$

Using,

$$BOD_t = L (1 - e^{(-kt)})$$

For $t = 5$ days and $k_{20} = 0.1$,

$$\begin{aligned} BOD_5 (20) &= 201.16(1 - e^{(-0.1 \times 5)}) \\ &= 201.16(1 - e^{(-0.5)}) \\ &= 201.16(1 - 0.6065) \\ &= 201.16 \times 0.3935 \\ &= 79.15 \text{ mg/litre} \end{aligned}$$

6. **What are the different types of sewerage system? What are their advantage and disadvantage?**

Answer:

Separate and combined sewerage systems are the two common types of sewerage systems.

Separate system:

Separate system consists of two different sewers, one for convergence of sewage and another for rainwater.

It requires large areas since cleaning is not possible in small sewers.

If the rainfall is heavy and for short period of time and if sanitary sewage is required to be pumped, then separate system is provided.

Advantages:

- i. Load on treatment plant is reduced because waste type is taken separately.
- ii. Uniform process for purification.
- iii. Reduces the possibility of pollution due to overflow during excess rainfall.
- iv. Rainwater can be directly discharged into river.

Disadvantages:

- i. Requires two set of sewers, hence costly.
- ii. Occupies large area.
- iii. Choking problem can be seen in separate system.
- iv. There is a possibility of wrong connection due to confusion.

Combined system:

Combined system consists of only one sewer to carry both sanitary sewage and storm water.

Sewage must be treated before discharged into the streams.

It is suitable if street washings are as impure as sewage and must be treated before discharging.

Advantages:

- i. Combined system reduces the cost as it requires only one sewer.
- ii. It also reduces maintenance cost.
- iii. Strength of sewage is reduced by dilution.
- iv. House plumbing is easy and economic.
- v. Self-cleansing velocity of sewer can be easily achieved in combined sewerage system.

Disadvantages:

- i. Large dimensions of sewer are required in combined system.
- ii. Ventilation problem may occur.
- iii. Load on treatment plant increases in combined system.
- iv. Overflow of sewage during rainy season can occur in combined system.
- v. Handling and transportation of large size of sewer is difficult.

7. **Suggest which one is best suited for Kathmandu valley and why?**

Also list the logic behind your selection of the particular type of treatment.

Answer:

The separate sewerage system is best suited for Kathmandu Valley, particularly for densely populated and rapidly urbanizing areas.

In a separate system, domestic and industrial wastewater is conveyed through sanitary sewers, while stormwater is carried separately through natural drains and stormwater drainage systems. It is suitable because of the following reasons:

- i. Kathmandu Valley receives intense rainfall during the monsoon. If stormwater and sewage are carried through the same sewer, the required sewer capacity becomes very large but a separate system avoids this problem.
- ii. Sanitary sewers carry only wastewater, so their design discharge is much smaller than that of combined sewers. Therefore, smaller and more economical sewer pipes can be used.
- iii. During heavy rainfall, a combined sewer can become overloaded and cause sewage overflow into rivers and public areas. A separate system reduces this risk.

The selection of a particular type of water or wastewater treatment is primarily governed by the quality of the raw water source, the geographic region, and economic feasibility. Based on the sources, the key considerations for selecting treatment types include:

i. Quality of water source:

- a. **Physical Impurities:** If water has high turbidity or suspended solids (common in river sources), sedimentation and filtration are mandatory.
- b. **Chemical Impurities:**
 - **Hardness:** If calcium and magnesium ions are present, softening (like the lime-soda process) is selected.
 - **Dissolved Gases/Metals:** For groundwater containing iron, manganese, or gases like CO₂, aeration is prioritized.
 - **Arsenic:** In specific Terai belts where arsenic levels are high, specialized removal units are required.
- c. **Biological Pathogens:** Every scheme intended for drinking must include disinfection to kill bacteria and viruses.

ii. Geographical conditions:

- **Hilly and Mountainous Regions:** Sources like pure perennial springs may require no treatment plant at all, as the water is relatively free from harmful materials.
- **Terai and Urban Areas:** Schemes using river water or large-scale groundwater in these flat regions must incorporate treatment plants to manage high suspended solids and potential surface contamination

iii. Cost-Effectiveness: The selected method should remove impurities economically. For example, Slow Sand Filters are often preferred for their high bacterial removal efficiency and lower operational complexity.

iv. Energy Requirements: The availability of electricity for pumping or sophisticated treatment (like activated sludge processes) influences whether a project opts for high-tech reactors or lower-energy solutions like oxidation ponds.

8. **What do you understand by community participation in water and sanitation sector? Why user committee is necessary in implementation, operation and maintenance of water supply sanitation projects. What are the critical features of community management in water and sanitary sector?**

Answer:

Community participation in the water and sanitation sector is a process where local beneficiaries, community organizations, and stakeholders actively engage in every stage of a project, including planning, design, implementation, operation, maintenance, and monitoring. This participatory approach ensures that the infrastructure developed meets the actual needs and priorities of the community, fostering long-term utilization and sustainability.

The formation of a Water Users and Sanitation Committee (WUSC) or User Committee is considered essential in implementation, operation and maintenance of water supply sanitation projects because of several reasons:

- i. **Sense of Ownership:** Early and active involvement creates a deep sense of ownership among users, making them more likely to protect and contribute to the system's longevity.
- ii. **Conflict Resolution:** Committees facilitate community consensus on project sites and canal/pipeline alignments, which minimizes future disputes regarding land acquisition, right-of-way, and water source rights.

- iii. **Resource Mobilization:** User committees effectively mobilize local resources, such as voluntary labor, cash contributions, and locally available raw materials, which significantly reduces construction and maintenance costs.
- iv. **Efficient Operation and Maintenance (O&M):** Local management allows for quick repairs of leaks or faults that might otherwise be delayed if managed by a distant government agency.
- v. **Indigenous Knowledge:** Integrating local farmers' knowledge of the terrain, soil, and river behavior leads to more pragmatic and cost-effective technological solutions.
- vi. **Sustainability and Transparency:** Committees are responsible for collecting tariffs and managing bank accounts, ensuring revenue is available for O&M while promoting financial transparency and accountability.

Community management refers to the process in which local people take an active role in the planning, implementation, operation, maintenance and management of water supply and sanitation facilities.

The critical features of community management are:

- i. **Community ownership:**
The community should develop a sense of ownership and responsibility towards water supply and sanitation facilities.
- ii. **Active participation:**
Local people should participate in planning, decision-making, implementation, operation and maintenance of the project.
- iii. **Inclusive representation:**
Women, disadvantaged groups, marginalized communities and other beneficiaries should have fair representation in decision-making.
- iv. **Effective operation and maintenance:**
The community should be capable of carrying out regular inspection, minor repairs and routine maintenance of the system.
- v. **Transparency and accountability:**
Decisions, financial transactions and project activities should be transparent and accountable to the users.
- vi. **Financial sustainability:**
Proper systems for tariff collection, budgeting, accounting and management of funds should be established to cover operation and maintenance costs.

9. **a. Discuss in briefly how global warming is affecting water supply sector in Nepal and what will be the proper remedial measures?**
b. As per environment protection act/ Regulation of Government of Nepal, discuss in brief the criteria for conducting IEE/EIA for development projects.

Answer:

a. Global warming denotes to the long-term increase in the average temperature of the Earth's atmosphere due to the increasing concentration of greenhouse gases. In Nepal, it is affecting the availability, quality, reliability and sustainability of water supply systems.

Effects of global warming:

- i. **Reduction in water availability:**
Changes in rainfall patterns, prolonged dry periods and increasing temperatures can reduce the availability of water from springs, streams and other sources, particularly in the hills and mountains.

- ii. **Drying of springs:**
Many traditional springs are becoming seasonal or drying up due to changes in rainfall and groundwater recharge, creating difficulties for rural water supply.
- iii. **Changes in precipitation pattern:**
Rainfall is becoming more irregular, with intense rainfall occurring over short periods and longer dry periods between rainfall events. This makes water-source planning more difficult.
- iv. **Floods and landslides:**
Intense rainfall increases floods, landslides and debris flows, which can damage intakes, pipelines, reservoirs and treatment plants.
- v. **Glacial lake outburst floods (GLOFs):**
Rising temperatures accelerate glacier and snow melt, increasing the risk of GLOFs in the Himalayan region. Such events can damage downstream water supply infrastructure.
- vi. **Deterioration of water quality:**
Floods and increased surface runoff can carry sediments, pathogens, agricultural chemicals and other pollutants into water sources, increasing turbidity and contamination.

Proper remedial measures include:

- i. **Conservation of water sources:**
Springs, watersheds, wetlands and recharge areas should be protected through watershed management, afforestation and controlled land use.
- ii. **Rainwater harvesting:**
Rainwater harvesting should be promoted for household, institutional and community-level water supply, particularly in water-scarce areas.
- iii. **Artificial groundwater recharge:**
Recharge ponds, infiltration trenches, recharge wells and other suitable techniques should be adopted to improve groundwater availability.
- iv. **Diversification of water sources:**
Dependence on a single source should be reduced by combining springs, groundwater, surface water and rainwater harvesting where feasible.
- v. **Climate-resilient infrastructure:**
Water supply infrastructure should be designed considering future floods, landslides, droughts, changing rainfall patterns and other climate-related hazards.
- vi. **Improved water storage:**
Appropriate reservoirs, storage tanks and small-scale water storage systems should be developed to store water during periods of surplus for use during dry periods.

b. Under the Environment Protection Act, 2019 (2076) and the Environment Protection Rules, 2020 (2077), development projects in Nepal must undergo an environmental study to identify and mitigate potential adverse impacts on natural, cultural, and social systems.

The criteria for determining whether a project requires a Brief Environmental Study (BES), an Initial Environmental Examination (IEE), or a full Environmental Impact Assessment (EIA) are primarily based on the scale, nature, and location of the project as prescribed in the schedules of the Regulations.

Types of environmental study:

- i. **Brief Environmental Study (BES):** Conducted for smaller proposals to identify basic mitigation measures for adverse effects.
- ii. **Initial Environmental Examination (IEE):** An analytical study conducted for medium-scale projects to ascertain if they will have significant adverse impacts.

- iii. **Environmental Impact Assessment (EIA):** A detailed study and evaluation required for large-scale projects likely to have substantial environmental consequences.

The Regulations categorize projects into Schedules 1, 2, and 3 based on specific technical criteria:

i. **Road Sector:**

- **IEE:** Required for the construction of new roads up to 25 kilometres long, tunnels between 1 and 3 kilometres, or cable cars between 1 and 5 kilometres.
- **EIA:** Required for new roads exceeding 25 kilometres, tunnels longer than 3 kilometres, or cable cars longer than 5 kilometres.

ii. **Energy and Hydropower:**

- **IEE:** Required for hydroelectricity projects with a capacity of 1 to 50 MW.
- **EIA:** Required for hydroelectricity projects with a capacity exceeding 50 MW.

iii. **Drinking Water and Irrigation:**

- **IEE:** Required for water supply projects serving 50,000 to 200,000 people or irrigation projects covering 200 to 2,000 hectares in the Terai.
- **EIA:** Required for water supply projects serving over 200,000 people or irrigation projects exceeding 2,000 hectares in the Terai.

iv. **Housing and Building:**

- **IEE:** Required for buildings with 7 to 10 floors or a height of 15.5 to 25 meters.
- **EIA:** Required for buildings with more than 16 floors or exceeding 50 meters in height.

Beyond technical dimensions, a project's social and environmental sensitivity can trigger higher-level studies:

- **Physical Displacement:** A project requires an IEE if it involves displacing 25 to 100 people. An EIA is mandatory if the displacement exceeds 100 people.
- **Protected Areas:** Proposals implemented within National Parks, Wildlife Reserves, or Conservation Areas generally require more rigorous studies (IEE or EIA) regardless of their size.
- **National Heritage:** While projects impacting national heritage sites generally require study, the Act specifies that reconstructing an ancient monument as it originally was does not require an environmental study report.

10. **Design a septic tank for a hotel of any school; where 125 residential students regularly living in school. Also design the soil absorption system for the disposal of the septic tank effluent percolation rate as 20 minutes per cm.**

Answer:

Given,

Number of residential students, $P = 125$

Assume water demand = 100 litres/capita/day

Assume sewage flow = 80% of water demand

Percolation rate of soil = 20 minutes/cm

Assume detention period = 24 hours

Assume liquid depth of septic tank = 1.5 m

Assume freeboard = 0.3 m

Assume sludge accumulation = 0.03 m³/person/year

Assume cleaning interval = 1 year

For the design of septic tank:

Water demand = 125 × 100 = 12,500 litres/day

Sewage flow = 80% of water demand

$$= \frac{80 \times 12,500}{100}$$

$$= 10,000 \text{ litres/day}$$

$$= 10 \text{ m}^3/\text{day}$$

Taking detention period = 24 hours,

Required liquid volume = Daily sewage flow × Detention period

$$= 10 \times 1$$

$$= 10 \text{ m}^3$$

Sludge accumulation = 0.03 m³/person/year

Therefore,

Sludge volume = 125 × 0.03 × 1

$$= 3.75 \text{ m}^3$$

Total volume = Liquid volume + Sludge storage volume

$$= 10 + 3.75$$

$$= 13.75 \text{ m}^3$$

Provide a septic tank of approximately:

Required volume = 14 m³

Assume width of tank, B = 2.0 m

Assume liquid depth, D = 1.5 m

Therefore,

Length, L = Required volume / (B × D)

$$= \frac{14}{2 \times 1.5}$$

$$= 4.67 \text{ m}$$

Adopt:

Length = 4.7 m

Width = 2.0 m

Liquid depth = 1.5 m

Freeboard = 0.3 m

Total depth = 1.5 + 0.3

$$= 1.8 \text{ m}$$

Therefore, provide a septic tank of: $4.7 \text{ m} \times 2.0 \text{ m} \times 1.8 \text{ m}$

The septic tank is divided into two compartments. Adopt the first compartment as approximately 2/3 of the total length and the second compartment as approximately 1/3 of the total length.

Therefore,

$$\text{First compartment length} = 2/3 \times 4.7$$

$$= 3.13 \text{ m}$$

$$\text{Second compartment length} = 1/3 \times 4.7$$

$$= 1.57 \text{ m}$$

Hence, the septic tank may be provided with two compartments of approximately:

$$\text{First compartment} = 3.1 \text{ m}$$

$$\text{Second compartment} = 1.6 \text{ m}$$

For the design of soil absorption system:

Given: Percolation rate = 20 minutes/cm

Therefore,

$$\text{Infiltration rate} = 1 \text{ cm} / 20 \text{ min}$$

$$= 0.05 \text{ cm/min}$$

Converting into cm/day:

$$\text{Infiltration rate} = 0.05 \times 60 \times 24$$

$$= 72 \text{ cm/day}$$

$$= 0.72 \text{ m/day}$$

Therefore, infiltration capacity of soil is:

$$= 0.72 \text{ m}^3/\text{m}^2/\text{day}$$

$$= 720 \text{ litres/m}^2/\text{day}$$

Daily septic tank effluent = 10,000 litres/day

Required absorption area = Daily effluent / Infiltration capacity

$$= \frac{10,000}{720}$$

$$= 13.89 \text{ m}^2$$

Therefore,

$$\text{Required soil absorption area} = 13.89 \text{ m}^2$$

Provide approximately:

$$\text{Required absorption area} = 14 \text{ m}^2$$

$$\text{Assume trench width} = 1.0 \text{ m}$$

Therefore,

$$\text{Required trench length} = \text{Absorption area} / \text{Trench width}$$

$$= \frac{14}{1}$$

$$= 14 \text{ m}$$

Hence, provide an absorption trench of approximately:

$$\text{Length} = 14 \text{ m}$$

$$\text{Width} = 1.0 \text{ m}$$

If required, the 14 m length may be divided into two parallel trenches:

$$\text{Length of each trench} = 14/2$$

$$= 7 \text{ m}$$

Therefore, provide 2 absorption trenches, each 7 m long and 1 m wide.